



NOVA SCHOOL OF
BUSINESS & ECONOMICS

Hedge Funds

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Assignment 3: Factor Investing

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Strategy Overview and Factor Selection:

This model constructs a monthly rebalanced equity long/short portfolio focusing on constituents of the S&P 500. Each month, the algorithm allocates equally sized positions to the 20 strongest quantitative prospects on the long side and the 20 weakest on the short side. The simulation spans approximately 11 years, from January 2014 through December 2025. To ensure robust validation, the timeline is strictly bifurcated: the period through December 2020 serves as the in-sample training environment, while 2021 onward is reserved for out-of-sample testing. The signal generation framework relies on four fundamental metrics. Earnings Yield and Return on Equity are prescribed core factors. To complement these, Profit Margin and Operating Income Growth were integrated. Profit Margin acts as a stringent quality filter, isolating companies with durable competitive advantages that trailing valuations often fail to capture. Operating Income Growth introduces a fundamental momentum overlay, capturing companies experiencing accelerating business operations. Individually, Operating Income Growth and Return on Equity demonstrated the most stable risk-adjusted performance during the in-sample phase, a robustness that is reflected across the full-sample individual factor metrics (*Table 3*) and their cumulative growth trajectories (*Figure 1*).

Data Processing and Market Frictions:

A primary pitfall in quantitative backtesting is the introduction of foresight bias. To simulate practical information flow, all fundamental data extracted from corporate filings is subjected to a strict delay of three months, preventing the algorithm from trading on unreleased earnings. Furthermore, comparative market standardisation is applied. We align the universe by dropping incomplete data rows and applying winsorization to clamp the extreme 2% statistical tails, which prevents severe accounting anomalies from distorting the standard scores. To model market microstructure realities, a transaction cost of 5 basis points is applied to all portfolio turnover. To actively mitigate this frictional drag, the strategy employs a hysteresis threshold using a buffer of $N+6$. Empirical testing compared portfolios with and without this rule. The unbuffered simulations suffered massive capital decay from excessive trading. By applying the buffer to keep existing holdings unless decisively superior candidates emerged, the strategy preserved capital and salvaged overall profitability.

Portfolio Construction, Factor Interactions, and Performance:

Rather than relying on static allocations, the final composite utilises a dynamic risk parity framework. Capital is scaled inversely to historical risk using a rolling 36-month volatility window. This allows the model to continuously adapt to shifting market regimes, assigning heavier influence to stable metrics like Profit Margin while constraining highly volatile signals like Earnings Yield. These dynamic weights are multiplied against individual stock Z-scores to aggregate a final master conviction score. It is crucial to note why this Z-score aggregation was chosen over a pure Rank-Rank composite. Combining negatively correlated factors like Value and Growth using raw cross-sectional ranks forces high-conviction stocks into the mediocre middle. Academics refer to this structural flaw as factor cancellation. Z-scores, however, preserve the magnitude of extreme conviction, preventing the portfolio from diluting capital into unexceptional compromise stocks. During the in-sample period (2014–2020), the algorithm actively learns the historical factor volatilities to populate its initial 36-month rolling window. Because the model must build its empirical memory during these early years, the dynamic risk parity approach initially yielded conservative baseline metrics, though it successfully navigated the severe 2020 market crash to deliver a 5.9% annualized return and a 0.13 Sharpe ratio (*Table 1*). Once this algorithmic memory stabilised, the true efficacy of the model materialised out-of-sample (2021–present). Operating through subsequent inflation shocks and the 2023–2024 economic recovery, the dynamic model generated an 11.07% annualized return with a Sharpe of 0.61 (*Table 2*). While the strategy remains susceptible to broad cross-asset selloffs, the rolling risk parity framework (*Figure 4*) successfully smoothed the variance across business cycles, vastly outperforming both the equal-weight method (*Figure 2*) and pure rank-based alternatives (*Figure 3*).

Appendix:

Table 1: Combined strategies performance metrics for the in-sample period (2014-2020)

Strategy	Metric	Value
Equal Weight	Ann. Return	2.55%
	Ann. Volatility	23.57%
	Sharpe Ratio	0.11
	Max Drawdown	-56.92%
	Calmar Ratio	0.04
	% Positive Months	57.14%
Rank-Rank	Ann. Return	-6.43%
	Ann. Volatility	17.48%
	Sharpe Ratio	-0.37
	Max Drawdown	-59.52%
	Calmar Ratio	-0.11
	% Positive Months	52.38%
Risk Parity	Ann. Return	5.90%
	Ann. Volatility	47.13%
	Sharpe Ratio	0.13
	Max Drawdown	-71.22%
	Calmar Ratio	0.08
	% Positive Months	51.39%

Table 2: Combined strategies performance metrics for the out-of-sample period (2021-present)

Strategy	Metric	Value
Equal Weight	Ann. Return	10.19%
	Ann. Volatility	19.36%
	Sharpe Ratio	0.53
	Max Drawdown	-30.78%
	Calmar Ratio	0.33
	% Positive Months	64.41%
Rank-Rank	Ann. Return	5.46%
	Ann. Volatility	19.53%
	Sharpe Ratio	0.28
	Max Drawdown	-40.41%
	Calmar Ratio	0.14
	% Positive Months	66.10%
Risk Parity	Ann. Return	11.07%
	Ann. Volatility	18.03%
	Sharpe Ratio	0.61
	Max Drawdown	-32.05%
	Calmar Ratio	0.35
	% Positive Months	62.71%

Table 3: Individual factor performance metrics

Factor	Metric	Value
Earnings Yield	Ann. Return	5.89%
	Ann. Volatility	38.26%
	Sharpe Ratio	0.15
	Max Drawdown	-53.16%
	Calmar Ratio	0.11
	% Positive Months	48.25%
Return on Equity	Ann. Return	4.67%
	Ann. Volatility	30.72%
	Sharpe Ratio	0.15
	Max Drawdown	-69.19%
	Calmar Ratio	0.07
	% Positive Months	60.84%
Profit Margin	Ann. Return	2.96%
	Ann. Volatility	22.18%
	Sharpe Ratio	0.13
	Max Drawdown	-69.73%
	Calmar Ratio	0.04
	% Positive Months	58.74%
Op. Income Growth	Ann. Return	3.05%
	Ann. Volatility	17.93%
	Sharpe Ratio	0.17
	Max Drawdown	-53.67%
	Calmar Ratio	0.06
	% Positive Months	60.14%

Figure 1: Cumulative individual factor portfolio metrics

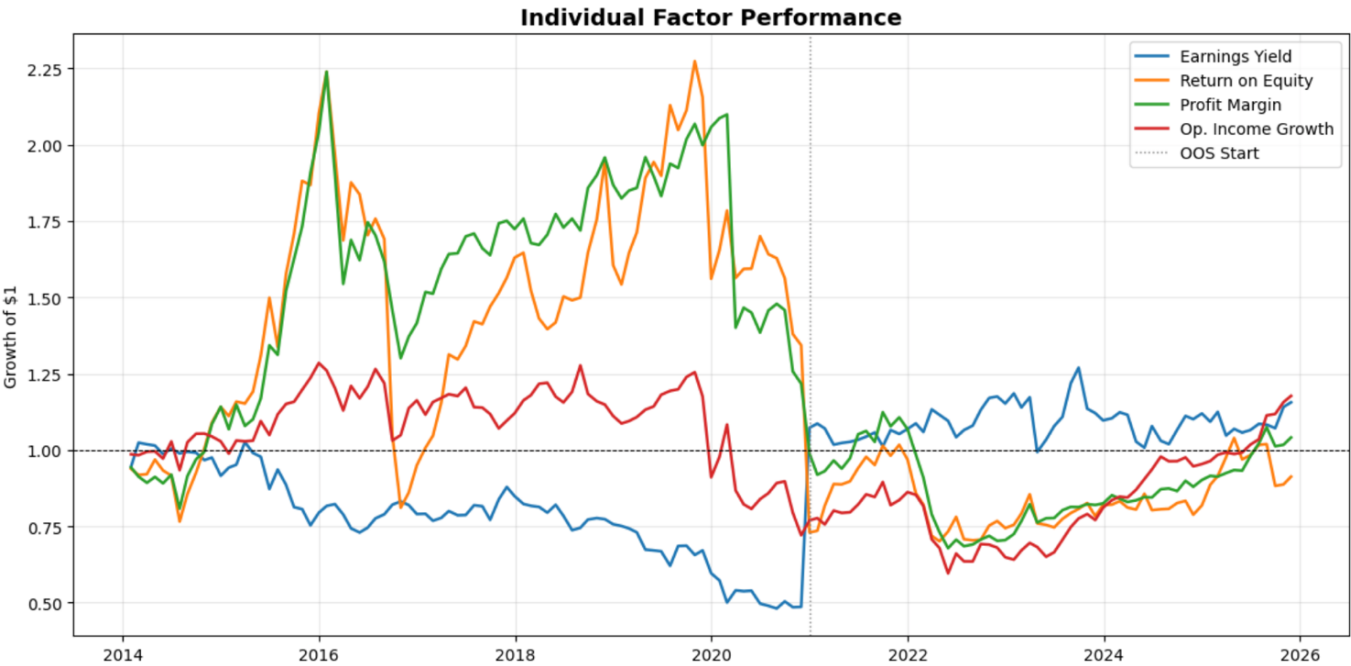


Figure 2: Equal-weight strategy portfolio metrics

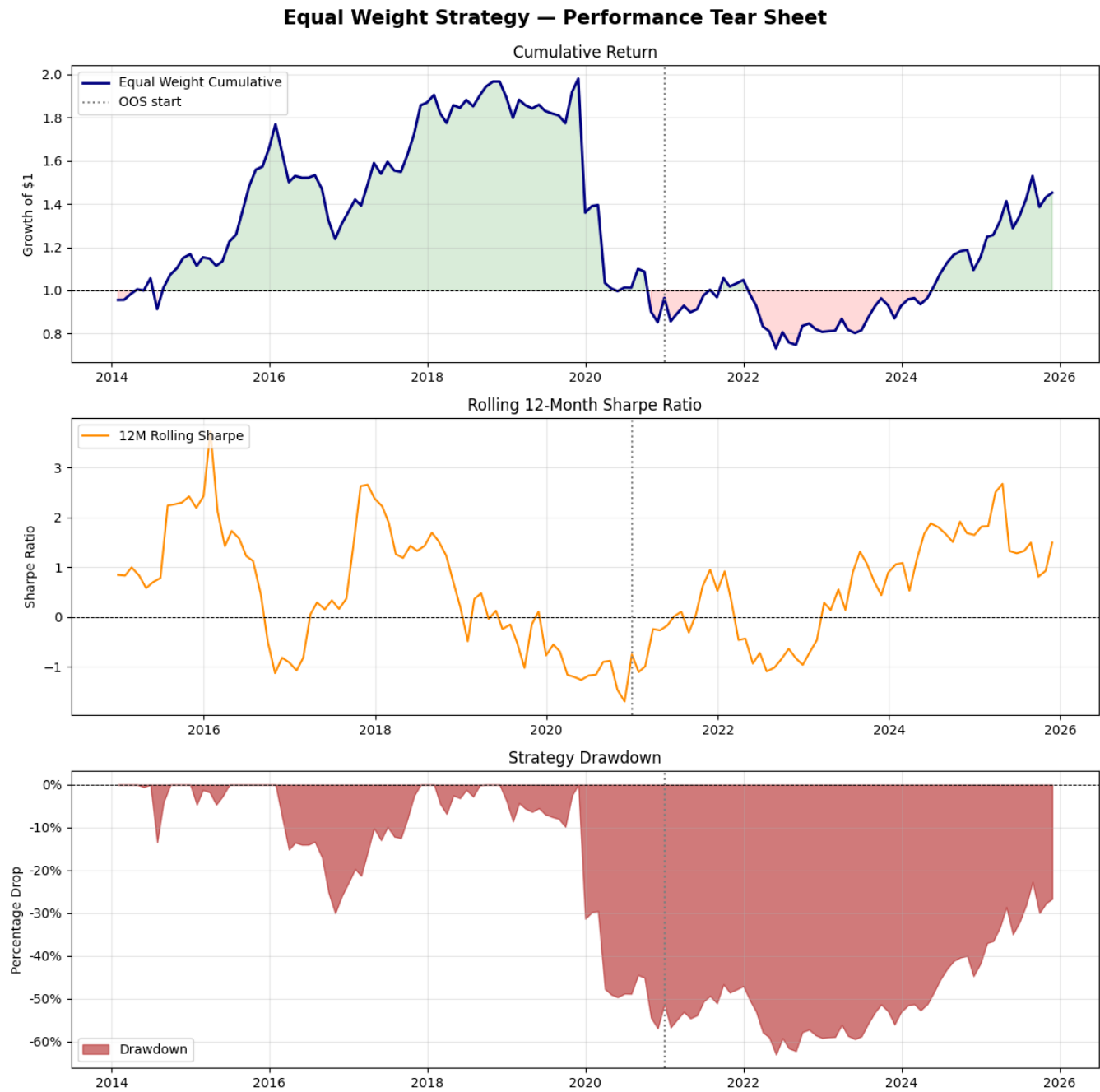


Figure 3: Rank-rank strategy portfolio metrics

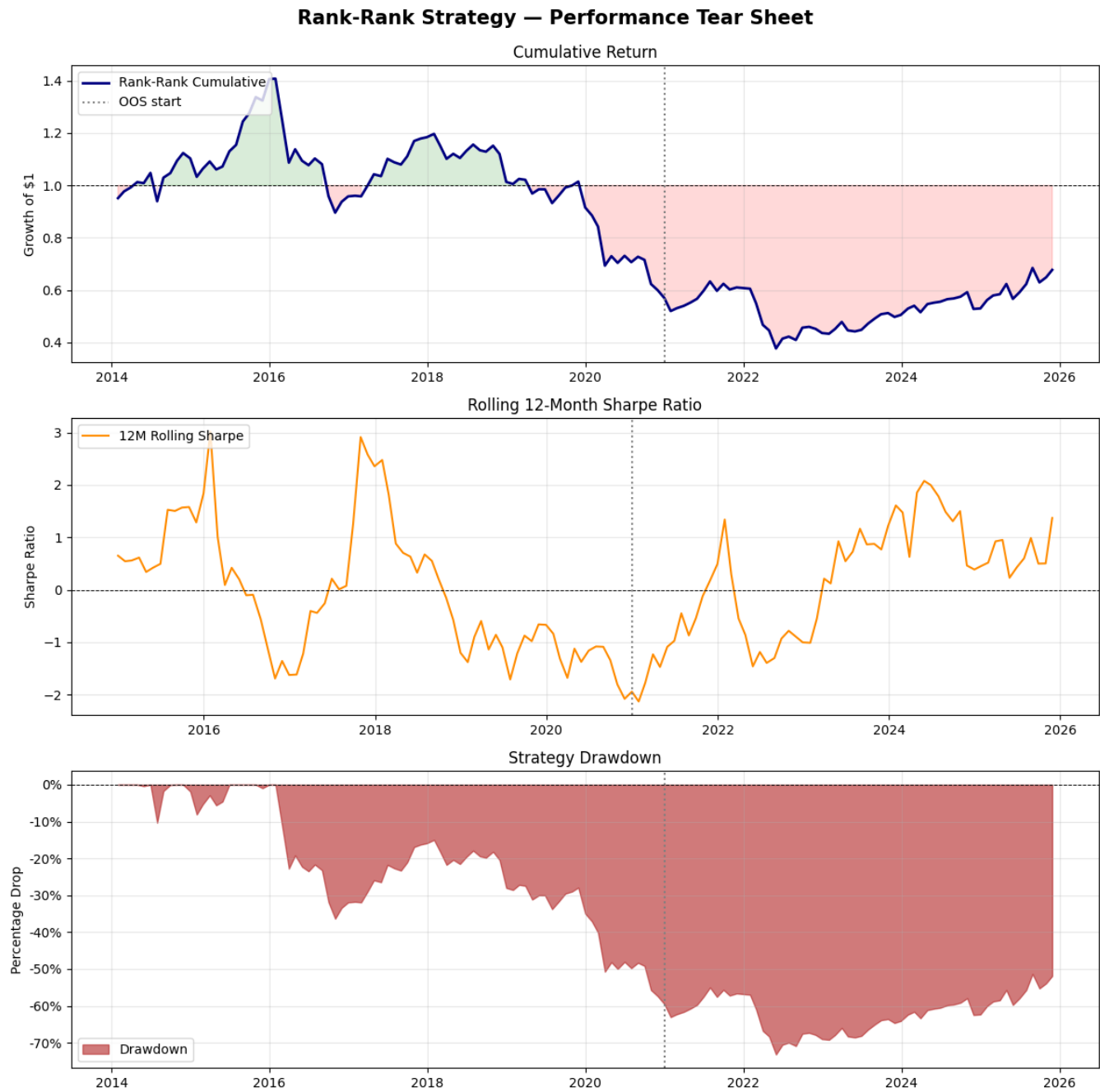


Figure 4: Risk parity strategy portfolio metrics

